TryHackMe - Windows Fundamental 1

Platform: TryHackMe
Room link: Windows Fundamentals 1

1. Introduction

Windows is one of the most widely used operating systems in personal computers and enterprise environments. Understanding its basic structure, user management, file system, and security features is essential for anyone learning cybersecurity.

This room introduces the Windows graphical user interface (GUI), file system, user accounts, permissions, User Account Control (UAC), and built-in administration tools.

The goal of this writeup is to explain the concepts and commands needed to explore Windows independently, without providing direct answers to the TryHackMe room questions.

2. Learning Objectives

By completing this room, you should understand:

- Different Windows editions and their features.
- The Windows desktop and graphical interface.
- The Windows file system and important directories.
- The purpose of the System32 directory.
- Windows user accounts and profiles.
- User permissions and access control.
- User Account Control (UAC).
- Windows Settings and Control Panel.
- Task Manager and its administrative features.

3. Tools and Environment

The room primarily uses a Windows virtual machine.

Useful built-in tools include:

Tool	Purpose
Command Prompt	Execute Windows commands
PowerShell	Run administrative commands and scripts
File Explorer	Browse files and folders
Settings	Configure Windows
Control Panel	Access traditional system configuration
Task Manager	Monitor processes and system performance
Computer Management	Manage users and system resources

4. Task-wise Walkthrough

Task 1 - Introduction to Windows

Windows is a graphical operating system developed by Microsoft.

An operating system manages computer hardware and software resources. It provides an interface through which users interact with applications, files, and system settings.

Important Windows components include:

- Kernel: Manages low-level system operations.
- File system: Organizes and stores data.
- User interface: Allows users to interact with the system.
- Device drivers: Enable communication between Windows and hardware.
- System services: Run background operations.

To inspect the installed Windows version, open Command Prompt and run:

```
winver
```

You can also use:

```
systeminfo
```

This displays information about the operating system, hardware, and system configuration.

Task 2 - Windows Editions

Microsoft provides different Windows editions for different users and environments.

Common editions include:

Edition	Typical use
Windows Home	Personal computers
Windows Pro	Business and professional environments
Windows Enterprise	Large organizations
Windows Education	Educational institutions

Some editions provide additional features, such as:

- BitLocker drive encryption.
- Group Policy management.
- Remote Desktop hosting.
- Advanced device management.
- Enterprise security controls.

To inspect the Windows edition from PowerShell:

```
Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion,
OsBuildNumber
```

Understanding the edition is useful because certain administrative and security features may not be available in every version.

Task 3 - The Desktop (GUI)

The Windows desktop is the primary graphical interface.

Important desktop components include:

- Start Menu: Provides access to installed applications and system options.
- Taskbar: Displays running applications and pinned shortcuts.
- Notification Area: Displays background application icons, network status, volume, and notifications.
- Desktop Icons: Provide shortcuts to files, folders, and applications.
- File Explorer: Allows users to browse the file system.

Useful keyboard shortcuts:

Shortcut	**Function**
Win + E	Open File Explorer
Win + R	Open Run dialog
Win + I	Open Settings
Win + D	Show desktop
Ctrl + Shift + Esc	Open Task Manager
Alt + Tab	Switch between applications

Practical exercise: Explore the Start Menu, taskbar, notification area, and desktop. Open File Explorer and navigate through different folders.

Task 4 - The File System

Windows commonly uses the NTFS file system.

NTFS stands for New Technology File System. It provides features such as:

- File and folder permissions.
- File compression.
- Disk quotas.
- Encryption support.
- Journaling and recovery capabilities.

Windows drives are represented using letters.

Examples:

```
C:\
D:\
```

E:\

A typical Windows directory structure looks like this:

```
C:\
│
├── Program Files
├── Program Files (x86)
├── Users
├── Windows
│   ├── System32
│   ├── Temp
│   └── Fonts
└── ProgramData
```

Important directories:

Directory	Purpose
C:\Windows	Core Windows operating system files
C:\Users	User profiles and personal data
C:\Program Files	Installed 64-bit applications, typically
C:\Program Files (x86)	Installed 32-bit applications on 64-bit Windows
C:\ProgramData	Shared application data
C:\Windows\Temp	Temporary system files

To display the current directory:

```
cd
```

To list files and folders:

```
dir
```

To navigate into a directory:

```
cd C:\Windows
```

To display hidden files:

```
dir /a
```

Practical exercise: Use File Explorer and Command Prompt to inspect the Windows directory structure. Avoid deleting or modifying system files.

Task 5 - Windows\System32

The System32 directory is one of the most important directories in Windows.

Its usual location is:

```
C:\Windows\System32
```

It contains essential operating system components, including:

- Windows system libraries.
- Executable files.
- Administrative utilities.
- Device-related components.
- System management tools.

Some important utilities found in or launched through this directory include:

Utility	**Purpose**
cmd.exe	Command Prompt
WindowsPowerShell	PowerShell environment
taskmgr.exe	Task Manager
services.msc	Manage Windows services
regedit.exe	Registry Editor
eventvwr.msc	Event Viewer

You can inspect the directory using:

```
cd C:\Windows\System32
dir
```

Security note: Malware sometimes uses filenames that resemble legitimate Windows system files. A familiar filename alone does not prove that a file is legitimate. Check its location, digital signature, and other properties.

Task 6 - User Accounts, Profiles, and Permissions

Windows supports multiple user accounts on a single computer.

Common account types include:

Administrator account

- Can perform system-level administrative tasks.
- Can install software and change important system settings.
- Can manage other accounts and permissions.

Standard user account

- Intended for everyday computer usage.
- Has limited administrative privileges.
- Cannot normally make protected system-level changes without elevation.

Windows also creates a profile for each user.

User profiles are commonly stored under:

C:\Users\

A profile may contain:

- Desktop
- Documents
- Downloads
- Pictures
- AppData

To identify the current user:

whoami

To inspect the current user's groups:

whoami /groups

To list local user accounts:

net user

To display the current user's profile directory:

echo %USERPROFILE%

File and folder permissions

NTFS uses access control lists (ACLs) to determine who can access files and folders. Common permissions include:

Permission	Description
Read	View file contents
Write	Modify file contents
Modify	Read, write, and modify files
Full Control	Includes permission management
Execute	Run executable files

To inspect permissions:

icacls C:\Users

Practical exercise: Explore your user profile and inspect permissions on a few folders. Do not change permissions on system directories.

Task 7 - User Account Control (UAC)

User Account Control is a Windows security feature designed to prevent unauthorized system changes.

When an application attempts to perform an administrative operation, Windows may display a UAC prompt.

UAC helps reduce the risk of unwanted changes by requiring user approval or administrator credentials when elevation is needed.

Examples of operations that may trigger UAC include:

- Installing software.
- Changing protected system settings.
- Modifying certain system directories.
- Running applications with elevated privileges.

Why UAC matters in cybersecurity

If malware runs without elevated privileges, UAC and Windows permissions can help limit what it can change. However, UAC is not a complete malware protection solution.

Avoid disabling UAC simply to make administrative tasks easier.

Task 8 - Settings and Control Panel

Windows provides two major interfaces for configuring the operating system.

Settings is the modern configuration interface.

It includes categories such as:

- System
- Devices
- Network & Internet
- Personalization
- Apps
- Accounts
- Privacy
- Update & Security

Open Settings using:

Win + I

You can also launch it through Run:

ms-settings:

Control Panel is the traditional Windows configuration interface.

It provides access to features such as:

- Programs and Features.
- Network and Sharing Center.
- User Accounts.

- System and Security.
- Administrative Tools.

Open Control Panel using:

```
control
```

Both interfaces are useful when investigating a Windows machine, especially because some settings are still easier to locate through Control Panel.

Task 9 - Task Manager

Task Manager is a built-in Windows utility for monitoring and managing running applications and processes.

Open Task Manager using:

```
Ctrl + Shift + Esc
```

Important tabs include:

Tab	Purpose
Processes	View running applications and processes
Performance	Monitor CPU, memory, disk, and network
App history	Review application resource usage
Startup	Manage startup applications
Users	View logged-in users and their resource usage
Details	Inspect individual processes
Services	View and manage Windows services

Task Manager is particularly useful during security investigations.

For example, an analyst may use it to:

- Identify suspicious processes.
- Investigate unusually high CPU or memory usage.
- Review applications configured to start automatically.
- Inspect running services.
- Identify applications that are not responding.

You can also launch Task Manager from Command Prompt:

```
taskmgr
```

Practical exercise: Explore the Processes, Performance, Startup, and Details tabs. Observe how applications and background processes appear.

5. Practical Exercises

Use the Windows virtual machine to perform these activities independently.

- Identify the Windows version and edition.

- Explore the desktop and Start Menu.
- Open File Explorer and inspect the C:\ drive.
- Navigate to C:\Windows\System32.
- Locate the current user's profile directory.
- Run whoami and inspect the output.
- Explore Windows user accounts.
- Inspect NTFS permissions using icacls.
- Understand when UAC prompts appear.
- Compare Windows Settings and Control Panel.
- Explore running processes using Task Manager.

6. Command Cheat Sheet

Command	Description
winver	Display Windows version
systeminfo	Display system information
whoami	Display current user
whoami /groups	Display user group memberships
net user	List local user accounts
echo %USERPROFILE%	Display user profile path
dir	List directory contents
dir /a	Include hidden files
cd	Change directory
icacls	Inspect file permissions
control	Open Control Panel
taskmgr	Open Task Manager
services.msc	Open Services management
eventvwr.msc	Open Event Viewer

7. Key Takeaways

After studying this room, you should understand:

a. Windows editions provide different features for different environments.
b. The desktop provides the primary graphical interface for interacting with Windows.
c. NTFS supports file permissions and access control.
d. The System32 directory contains important Windows components.
e. User accounts and profiles help organize access to a computer.
f. NTFS permissions determine what users can do with files and folders.
g. UAC helps protect the operating system from unauthorized administrative changes.
h. Settings and Control Panel provide access to system configuration.
i. Task Manager helps monitor applications, processes, and system resources.

8. Conclusion

Windows Fundamentals 1 introduces the basic structure and administration of the Windows operating system.

Understanding the file system, user accounts, permissions, UAC, and built-in management utilities provides a foundation for more advanced Windows security topics.

These concepts are particularly useful when performing Windows enumeration, investigating suspicious activity, and understanding how attackers interact with Windows systems.